Aquifer Sensitivity Dataset | Research Brief

Peace Engineering • Cuba, New Mexico • Dataset 3

Dataset and relevance to Cuba

The supplied CSV contains 613 polygon-attribute records with four fields: OBJECTID, SENSITIVITY, Shape__Area, and Shape__Length. Sensitivity classes are Moderate (Mod): 339, High: 152, and Less: 122. These are polygon counts—not percentages of land area. The CSV has no coordinates, polygon geometry, aquifer names, well IDs, dates, or water measurements.

Cuba-specific relevance: NOT ESTABLISHED. The file alone cannot show whether any mapped polygon overlaps Cuba, its municipal wells, or their source aquifer. A corresponding GIS layer and documentation of what "sensitivity" means are required. Do not describe this as evidence of Cuba's water depletion or contamination without that verification.

Research-question support

Question	Support	What this file can contribute
Q1 · Groundwater trends	None	No water-level observations or dates; cannot plot decline or recovery.
Q2 · Well/aquifer characteristics	None	No well depth, depth to water, or aquifer-unit identifiers.
Q3 · Cuba depletion	None	Sensitivity classes do not quantify water storage, recharge, or pumping.
Q5 · Water quality	Potential / indirect	Could contextualize vulnerability if metadata confirms a water-quality meaning; no measured chemistry.
Q8 · Monitoring gaps	Limited	Not a monitoring-well inventory; cannot identify monitoring coverage from this CSV.
Q9 · Stress dashboard	Conditional	Could supply a mapped sensitivity indicator once geometry, definition, and Cuba overlap are verified.

Recommended work for the report

1. Download the companion GIS polygon layer (e.g., GeoJSON or shapefile) and its metadata; determine what "High," "Mod," and "Less" actually measure. 2. Overlay the layer with Cuba, municipal wells, and aquifer boundaries. 3. If Cuba overlaps mapped polygons, create a map and compute sensitivity-class area shares using an appropriate projected coordinate system. 4. Use separate USGS/NMBGMR well data for groundwater trends and OSE records for wells and pumping.

Plotting caution

A bar chart of the 613 records by sensitivity class is possible, but it describes the downloaded polygon inventory—not Cuba. Shape__Area and Shape__Length are GIS-derived values with unspecified units/coordinate reference system in this CSV; do not label them acres, square miles, or feet without metadata.

source: https://catalog.newmexicowaterdata.org/dataset/aquifer-sensitivity

source: corresponding geographic map https://catalog.newmexicowaterdata.org/dataset/aquifer-sensitivity/resource/edf278aa-7c5b-4b65-97aa-c6825fa9015c